

Vigneshwaran D

Senior Manager – Enterprise AI

Enterprise AI Platform Architecture | Agentic AI | AI Platform Strategy | Multi-Agent Systems

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PROFESSIONAL SUMMARY

Senior Manager – Enterprise AI with 14+ years of experience leading enterprise AI platform engineering, digital transformation, and intelligent automation initiatives across Healthcare, BFSI, and Retail. Over 5+ years of expertise designing and delivering production-grade Generative AI, Agentic AI, Multi-Agent Systems, LLM, and RAG solutions on Azure. **Architect and Primary Inventor of two U.S. patent-filed AI platforms** — an Agentic Voice AI Call Training Simulator (App. No. 19/546,120) built on Azure OpenAI Realtime API, ElevenLabs, and HeyGen; and a Multi-Tenant Healthcare Mailbox Workflow Automation Platform (2026) with NLP-based PHI/PII detection. Specializes in AI Platform Strategy, Solution Architecture (HLD/LLD), AI Governance, LLMOps, and Model Evaluation, with a proven track record of building scalable enterprise AI platforms, mentoring engineering teams, and driving AI adoption that delivers measurable business value.

TECHNICAL SKILLS

LLM Engineering: GPT, Claude, LLaMA, Prompt Engineering, Fine-tuning (LoRA/QLoRA), RAG, Guardrails, Responsible AI, Structured Outputs, Function Calling

Agentic AI: CrewAI, LangGraph, AutoGen, MCP, A2A, Tool Calling

AI Platform Engineering: Azure AI Foundry, Azure ML, Prompt Flow, Semantic Kernel, LiteLLM

Enterprise Architecture: Solution Architecture (HLD/LLD), AI Platform Strategy, AI Governance, Microservices, REST APIs

Cloud & AI Platforms: Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure Functions, AWS, Docker, Kubernetes (K8s)

Data & Storage: PostgreSQL, MongoDB, Azure SQL, Pinecone, ChromaDB

Programming: Python, TypeScript, Node.js, FastAPI, React.js, SQL

Security & Compliance: Azure AD (MSAL), JWT, RBAC, Microsoft Presidio, Data Governance

Conversational AI: Azure OpenAI Realtime API, ElevenLabs, HeyGen

AI Evaluation & Observability: LangSmith, RAGAS, MLflow, BLEU, ROUGE

WORK EXPERIENCE

EXL Service — Senior Manager — Enterprise AI

Feb 2025 – Present

Leading Enterprise AI Platform Engineering and AI Transformation across Healthcare, Banking, and Insurance business units — owning AI Strategy, Platform Architecture, Governance, and Production Deployment.

- Defined Enterprise AI platform standards spanning architecture, model evaluation, security, and observability across multiple AI initiatives.
- Led AI solution architecture reviews (HLD/LLD validation) for enterprise GenAI platforms, and established reusable platform components including authentication, RBAC, and MCP services.
- Partnered with Product, Security, and Delivery leadership to define enterprise AI platform roadmap and adoption strategy.
- Mentored AI Engineers and Architects on Agentic AI, RAG architecture, and Azure AI best practices, standardizing prompt engineering and evaluation methodology across delivery teams.

AI Training Simulator - Patent Filed — App. No. 19/546,120

- Invented and architected a patent-pending voice AI call training platform — a distributed 5-service system (React 19, MCP AI server, CrewAI scoring service) supporting 5,000+ concurrent users at sub-200ms p99 API latency.
- Engineered real-time voice-to-voice call simulation integrating Azure OpenAI Realtime API, ElevenLabs, and HeyGen live avatar — reducing call coach headcount by 60% across 3 enterprise verticals.
- Built a 5-agent evaluation pipeline (CrewAI + LiteLLM) scoring 10,000+ monthly calls across 4 quality dimensions — eliminating 100% of manual QA review, fully traced in LangSmith and benchmarked with RAGAS.
- Delivered a 9-layer enterprise security architecture (JWT, Azure AD SSO/MSAL, RBAC, audit logging) achieving compliance for regulated BFSI and Insurance clients, deployed on Azure with 99.9% uptime SLA.

Mailbox Workflow Automation Platform - Patent Application Filed (2026)

- Designed and built a full-stack multi-tenant healthcare email automation platform (React 18, TypeScript, Node.js, PostgreSQL + Drizzle ORM) converting incoming emails into structured incidents with SLA-driven routing and real-time status tracking — reducing manual triage effort by an estimated 80%.
- Integrated Microsoft Presidio NLP engine for server-side PHI/PII detection and client-side masking utilities — enabling HIPAA-adjacent compliance for healthcare email workflows at scale.
- Architected full multi-tenancy with scoped data isolation, fine-grained RBAC, and Azure AD SSO — deployed as a Progressive Web App (PWA) with offline access support.

PepsiCo — Manager — Digital Assistant (AI/ML)

Oct 2023 – Feb 2025

Led Enterprise AI Architecture and AI Platform Engineering initiatives supporting global operations across Procurement, Finance, Supply Chain, and Marketing.

- Defined architecture standards and led technical solution governance, including architecture reviews, for enterprise GenAI applications.
- Led model evaluation strategy using ROUGE, BLEU, and human benchmark validation, and standardized RAG implementation patterns across multiple AI solutions.
- Collaborated with executive stakeholders to prioritize AI initiatives and define the implementation roadmap.

Intelligent Document Summarization (SuMaaS)

- Reduced research operations cost by **\$900K annually** (60% headcount reduction, **40→16 analysts**) by engineering a RAPTOR + MapReduce + Refine document summarization pipeline supporting 10+ file formats and processing 1,000+ documents/week with LLM-based abstractive summaries.
- Evaluated and benchmarked summarization quality using **ROUGE and BLEU metrics** against human baselines; selected optimal chunking strategy (RAPTOR vs. Stuff vs. MapReduce) per document type through systematic ablation.

BI Chatbot — SQL RAG & Predictive Analytics

- Eliminated **7 FTE MIS positions** (\$350K+ annual savings) while reducing data query response time by **60%** by building a LangChain multi-agent chatbot that translates natural language to SQL, executes against live databases, generates AI-driven charts, and surfaces next-best-action predictions.
- Implemented SQL RAG with schema-aware custom agents — enabling non-technical users to query complex multi-table databases with **90%+ query accuracy** validated against a golden test set.

AI Headline & Tagline Generator for Social Media

- Built a multi-agent LLM pipeline generating campaign copy for product launches, seasonal promotions, and brand campaigns — with configurable prompts per marketing objective and audience demographic targeting.

Managed Conversational AI Platform Engineering, RPA Architecture, and Enterprise Automation initiatives while leading cross-functional engineering teams.

- Reduced admin team headcount from 10 to 2 FTEs by building an AWS Lex hyper-automation chatbot handling password resets, account creation, and login issues — eliminating 80% of manual administrative workload.
- Led end-to-end development and production deployment of Zenni Chatbot (Order Status, Offers, Product Suggestions) using Power Virtual Agent, managing team, stakeholder alignment, and post-launch support.
- Designed a custom bot orchestrator for multi-language automation deployment — replacing UiPath licensing costs while providing version control, scheduling, and automated execution.

AGS Health — Supervisor — RPA Architect*Mar 2012 – Sep 2020*

Established enterprise RPA platform architecture supporting 165+ production automations.

- Designed and built a custom Python + MongoDB RPA orchestrator managing 165+ production bots across 14 sub-processes with 85% active uptime — replacing equivalent UiPath orchestrator functionality at a fraction of licensing cost.
- Delivered end-to-end healthcare process automation using Selenium, UiPath, Blue Prism, and custom scripting — cutting average manual processing time by 70% across 50+ business processes and reducing error rates through automated quality gates.

PATENTS & INVENTIONS

- AI Call Training Simulator — **U.S. Patent App. No. 19/546,120** (Filed 2025). Novel architecture for real-time voice AI simulation of customer service scenarios using multi-agent LLM scoring and live avatar synthesis; first system to combine Azure OpenAI Realtime API + ElevenLabs + HeyGen for enterprise training at scale.
- Mailbox Workflow Automation Platform — Patent application filed and in progress (2026). Multi-tenant healthcare email triage system with integrated NLP-based PHI/PII detection and SLA-driven incident routing.

AWARDS & RECOGNITION

- Best AI Solution 2025 – EXL Service (2025)
- "Act As Owners" & "Celebrate Success" — PepsiCo (2024)
- Best Performer Award — AGS Health (2017, 2019, 2020)

EDUCATION

Bachelor of Computer Applications (BCA)*May 2011*

Sathyabama Institute of Science and Technology, Chennai | Distinction — 75.6%

CERTIFICATIONS

- **AWS Certified Solutions Architect Associate (SAA-C03)** — (Feb 2022)
- **Virtual Agent Development with Dialogflow CX** — (Apr 2023)
- **RPA Infrastructure Engineer** — UiPath (Apr 2022)
- **UiPath Security** — UiPath (Mar 2022)